Herald

Herald — Telegram Integration (Operator Guide)

Field	Value
Revision	1
Created	2026-05-28
Last modified	2026-05-28
Status	active
Status summary	Operator-facing deep-dive for Herald's Telegram integration — Herald's most-used messenger channel, live since Wave 6 (HRD-011 outbound substrate + HRD-100 inbound runtime). Documents the two-way model (inbound `pherald listen` → `getUpdates` long-poll → Claude Code dispatch → outbound reply via `bot.Send` / `bot.SendReply`), the BotFather walkthrough (bot creation, privacy-mode toggling), every required env var with concrete acquisition steps, the end-to-end verification recipes (`getMe`, `getChat`, hermetic `go test`, `pherald listen` smoke), Bot API hard limits (4096-char text, 50MiB documents, 10MiB photos free / 50MiB premium), the per-channel content-addressed inbox (`~/.herald/inbox/tgram/<sha256>.<ext>`), chat-type taxonomy (private / group / supergroup / channel — what `chat_id` looks like for each), reply threading + attachment fan-out (the M3 rationale for serialised replies over multi-attachment albums), the self-filter that prevents echo loops (`bot.Me.Username` cached at Subscribe boot — fail-loud if empty), a 10-entry troubleshooting cookbook (401/403/409/413/429 + privacy-mode + webhook conflicts + zero-byte downloads), the operator pre-deploy audit checklist, and the full cross-reference index to spec V3, HRDs, source files, and sibling guides.
Issues	none
Issues summary	—
Fixed	(n/a — new guide)
Continuation	bump when MTProto user-client lands (Wave 7+ — current bot-to-bot wall workaround captured in `feedback` memory `telegram_bot_to_bot_wall`); bump when local Bot API server support lands (lifts the 50MiB document cap); bump when forum-topic threading is rounded out beyond V1 stub; bump when interactive call routing (Capabilities.InteractiveCall) lands; bump when webhook ingress (Step 7 in `messengers/TELEGRAM.md`) is implemented.

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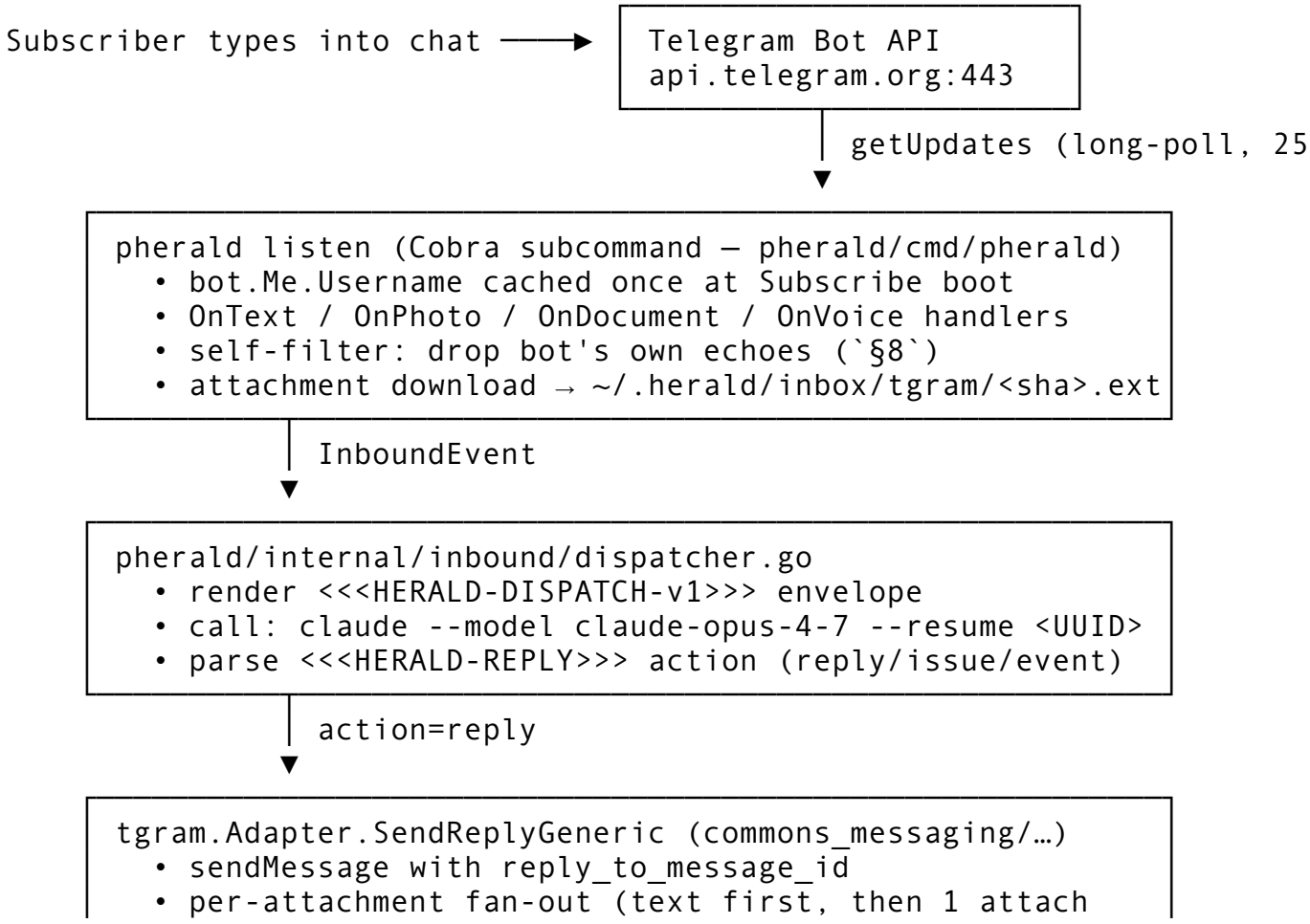
§1. What is Herald's Telegram integration?

Telegram is Herald's reference messenger — the first channel that went live (Wave 6, 2026-05-21) and the channel against which every cross-cutting invariant in the messaging plane was first proven. It is bidirectional: Herald can both deliver outbound notifications to a chat AND react to inbound messages a subscriber posts in that same chat.

§1.1 Two halves of one channel



```
    per reply — M3 rationale, §7)
```

The outbound half (`Send` / `SendReply` / `SendReplyGeneric`) is available to ANY Herald flavor binary — `pherald serve`, `sherald`, `cherald`, `bherald`, `rherald`, `iherald`, `scherald`, `qaherald` — through the `commons.Channel` outbound contract (spec §11.0). The inbound half is exclusive to `pherald listen` (and, by extension, `qaherald` which embeds the same inbound runtime for QA round-trips).

§1.2 Status

- **Outbound** — LIVE since Wave 6 (HRD-011). Hermetic tests + Wave 6.5 paired §1.1 mutation gate pin the wire-byte shape. Live evidence: `docs/qa/HRD-011-LIVE-*/` (operator-supplied).
- **Inbound** — LIVE since Wave 6 (HRD-100). `pherald listen` Cobra subcommand drives the closed-loop runtime; T10a `--qa-out-dir` JSONL journaling captures every dispatch. Live evidence: `docs/qa/HRD-100-LIVE-*/` (operator-supplied; T10b currently open until live transcripts land).
- **Multi-channel fan-in** — LIVE since Wave 7 (HRD-114). Telegram now coexists with Slack under one shared inbound dispatcher; see `MESSENGER_CHANNELS.md`.

§1.3 Where the code lives

- `commons_messaging/channels/tgram/tgram.go` — `Adapter` struct, `New` / `NewWithCreds` constructors, `init()` registry binding (Wave 7 T2), `BotSelfIdentity` / `SendReplyGeneric` / `DownloadAttachment` channel-interface methods.
- `commons_messaging/channels/tgram/send.go` — outbound `Send` + native `SendReply(ctx, chatID, body, replyToID, attachments)`.
- `commons_messaging/channels/tgram/subscribe.go` — inbound `Subscribe(ctx, h)` long-poll loop, `stampAndIsSelfEcho` filter, OnText/OnPhoto/OnDocument/OnVoice handlers.
- `commons_messaging/channels/tgram/attachments.go` — `DownloadAttachment(ctx, bot, fileID, mime)` → content-addressed write into `~/.herald/inbox/tgram/<sha>.<ext>`.
- `commons_messaging/channels/tgram/healthcheck.go` — `HealthCheck` (calls `getMe`).
- `commons_messaging/channels/tgram/persist.go` — outbound-delivery-evidence persistence into the `outbound_delivery_evidence` Postgres table (RLS-aware via `SendForTenant`).
- `pherald/cmd/pherald/listen.go` — the `pherald listen` Cobra subcommand that wires the inbound runtime.
- `pherald/internal/inbound/dispatcher.go` — Claude Code dispatch + `<<<HERALD-REPLY>>>` action router.

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§2. BotFather walkthrough (creating a bot)

A Telegram **bot** is a user-like entity owned by your real Telegram account. You create it by chatting with `@BotFather`, Telegram's official bot-management bot.

§2.1 Create the bot

1. Open Telegram (mobile, desktop, or web).
2. Tap the search icon and search `BotFather`.
3. Open the chat with the verified-blue-checkmark account `@BotFather` — DO NOT pick any impostor `BotFatherSomething` accounts.
4. Send `/start` (one-time intro).
5. Send `/newbot`.
6. BotFather replies asking for the bot's **display name**. Pick a human-readable string (any UTF-8). Examples: `Herald Ops`, `My Project CI`, `Acme Notifications`.
7. BotFather then asks for the bot's **username**. Must be globally unique on Telegram and MUST end with `bot` or `_bot`. Examples: `herald_ops_bot`, `myproject_ci_bot`, `acme_notify_bot`.
8. BotFather replies with a token of the canonical form:

   ```
   1234567890:AAH1lab2c3d4e5f6g7h8i9j0kK_LmN3oPqR4S5t
   ```

   The `1234567890` prefix is the bot's numeric user id; the `AAH…` portion is the HMAC the Bot API uses to authenticate API calls. Treat the entire string as a password.
9. **Copy the token into your password manager immediately** — BotFather will display it once and only show subsequent reads on demand via `/mybots → <select bot> → API Token`.

§2.2 Privacy mode

By default, every new Telegram bot is created with **privacy mode ENABLED**. With privacy on, a bot in a group can see ONLY (a) commands directed at it (e.g. `/foo`), (b) messages that mention it by `@username`, and (c) replies to its own messages. Regular group chatter is invisible to the bot's `getUpdates`.

For Herald, the standard Wave 6 use-case is "DM with the bot, the bot relays the conversation to Claude Code" — which works fine with privacy mode ON (private DMs are always fully visible to the bot regardless of the privacy setting). But if you want the bot to live in a group and observe ALL traffic, you MUST disable privacy:

1. DM `@BotFather`.
2. Send `/setprivacy`.
3. Select your bot from the inline keyboard.
4. Choose `Disable`.

Privacy-mode-on is the single most common cause of "the bot doesn't see my group messages" support tickets. Always start with privacy-mode-off if the bot lives in a group.

§2.3 Add the bot to a chat

For a 1:1 DM (the simplest path — preferred for `pherald listen` smoke):

1. In Telegram, search for the bot's `@username`.
2. Tap **Start** to open a DM. The `/start` button (auto-injected by Telegram's client for first-contact with a bot) registers YOU as a chat the bot knows about.
3. Send any plain-text message (e.g. `hello`).

For a group / supergroup:

1. Open the group, tap the title to enter group settings → **Add Members**.
2. Search for your bot's `@username`, tap to add.
3. (Optional, recommended for non-admin bots in supergroups) Promote the bot to **Admin** so it can send messages (in some channel configurations bots must be admins to post — see §6 for `channel` chat type specifics).
4. Send a test message in the group.

For a channel (one-way broadcast — Herald sends, no inbound):

1. Open the channel, channel settings → **Administrators → Add Administrator**.
2. Add the bot. The channel administrator scopes (`Post Messages`, `Edit Messages of Others`, `Delete Messages of Others`) are sufficient — Herald does not need anything more.
3. (Channels do not generate `getUpdates` inbound events that flow through Herald's current path — inbound is private + group + supergroup only.)

§2.4 Optional BotFather knobs

Other commands you may want to set once via BotFather:

Command	Purpose
`/setname`	Change the display name.
`/setdescription`	Long-form bot description shown in the bot's profile page.
`/setabouttext`	One-line tagline shown next to the bot's name in the "share contact" UI.
`/setuserpic`	Upload an avatar (recommended — set the Herald logo).
`/setcommands`	Register `/foo`-style commands so the Telegram client renders an inline `/` menu. Herald does not currently require this.
`/deletebot`	Permanently delete the bot. Irreversible.
`/revoke`	Invalidate the current token + issue a fresh one. Use this immediately if the token leaks.

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§3. Required env vars + how to obtain each

Herald's Telegram channel reads exactly two env vars at boot:

Variable	Required for	Format	Example
`HERALD_TGRAM_BOT_TOKEN`	outbound + inbound	`<bot-id>:<api-token>` from BotFather	`1234567890:AAH1lab2c3d4e5f6g7h8i9j`
`HERALD_TGRAM_CHAT_ID`	outbound default destination	numeric (private = positive; group = negative;	`-1001234567890`

Variable	Required for	Format	Example
		supergroup = `-100…` prefix; channel = `-100…` prefix)	

Optional / reserved (NOT required for Wave 6/7 functionality):

Variable	Required for	Notes
`HERALD_TGRAM_LIVE_INBOUND`	integration-test gating	Set to `1` to enable live inbound integration tests in `commons_messaging/channels/tgram/subscribe_integration_test.go`.
`HERALD_TGRAM_WEBHOOK_SECRET`	future webhook ingress	Reserved for a future HRD; webhook ingress is currently NOT implemented. Long-poll via `getUpdates` is the only inbound transport.
`HERALD_INBOX_DIR`	attachment storage	Overrides the per-channel inbox parent dir (default: `~/.herald/inbox/`). Per-channel subdir name is fixed to `tgram/`.

§3.1 Obtaining `HERALD_TGRAM_BOT_TOKEN`

The bot token is the canonical string BotFather emits at `/newbot` (see §2.1 step 8). It is the SAME string returned by `/mybots → <bot> → API Token` on demand. If lost, rotate via `/revoke` — there is no recovery path.

§3.2 Obtaining `HERALD_TGRAM_CHAT_ID`

For a **1:1 DM** (positive integer):

1. Open the DM with your bot.
2. Send any plain-text message (e.g. `id-probe`).
3. From a shell with curl:

```
curl -s "https://api.telegram.org/bot${HERALD_TGRAM_BOT_TOKEN}/getUpdates" | jq '.result[].message.chat'
```

4. Read `.id` — that integer (positive for DMs) is your `HERALD_TGRAM_CHAT_ID`.

For a **group** (negative integer):

1. Add the bot to the group (§2.3).
2. Send any message in the group (e.g. `@your_bot_username id-probe` if privacy mode is ON — otherwise any plain text works).
3. Same `getUpdates` curl as above; the `chat.id` is a negative integer.

For a **supergroup** (negative integer starting with `-100`):

1. Same as group (Telegram auto-converts groups to supergroups when admin features are enabled).
2. The chat id begins with the literal prefix `-100` followed by the numeric body, e.g. `-1001234567890`.

For a **channel** (negative integer starting with `-100`):

1. Add the bot as channel admin (§2.3).
2. Post a message to the channel.
3. Same `getUpdates` curl; if the bot has post permissions it will receive the `channel_post` update with `.channel_post.chat.id`.

Helper utility bots (alternative to manual curl): add `@userinfobot` or `@RawDataBot` to your chat — they reply with the chat id as plain text. These are third-party bots; vet them before sharing sensitive chat content.

Herald-provided diagnostic (if installed): `bash scripts/tgram_diagnose.sh` is the canonical sanity script in this repo. It calls `getUpdates`, parses the response, and prints chat ids in human-friendly form. Use it during onboarding.

§3.3 Copy-pastable `.env` snippet

```
# Herald — Telegram (HRD-011 outbound + HRD-100 inbound)
export HERALD_TGRAM_BOT_TOKEN='1234567890:AAH1lab2c3d4e5f6g7h8i9j0kK_LmN3o
export HERALD_TGRAM_CHAT_ID='-1001234567890'

# Optional (defaults shown):
# export HERALD_INBOX_DIR="$HOME/.herald/inbox"
# export HERALD_TGRAM_LIVE_INBOUND='1'      # only to enable live
        integration tests
```

Per `OPERATOR_CREDENTIALS.md` §"Resolution order (12-factor)", exported shell vars take precedence over `.env`. Put production tokens in your shell rc; put dev/test tokens in `.env` (the project-local `.env` is gitignored).

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§4. Verifying the integration end-to-end

There are four progressive verification steps. Each is cheap; each catches a different class of misconfiguration. Run them in order before promoting a deploy.

§4.1 Pre-flight: probe the Bot API directly

The cheapest sanity check is a raw `getMe` curl — it proves the token is valid + the API endpoint is reachable + your network can talk to Telegram.

```
curl -s "https://api.telegram.org/bot${HERALD_TGRAM_BOT_TOKEN}/
        getMe" | jq
```

A healthy response looks like:

```
{
  "ok": true,
  "result": {
    "id": 1234567890,
    "is_bot": true,
    "first_name": "Herald Ops",
    "username": "herald_ops_bot",
    "can_join_groups": true,
    "can_read_all_group_messages": false,
    "supports_inline_queries": false
  }
}
```

Look for `ok: true` and `result.username` populated. If `ok: false`, the token is wrong or revoked (see §9.1). The `can_read_all_group_messages: false` field reports the privacy-mode setting — `true` means you successfully disabled privacy via `/setprivacy`.

§4.2 Verify the bot can see the target chat

```
curl -s "https://api.telegram.org/bot${HERALD_TGRAM_BOT_TOKEN}/
        getChat?chat_id=${HERALD_TGRAM_CHAT_ID}" | jq
```

Healthy response:

```
{
  "ok": true,
  "result": {
    "id": -1001234567890,
    "type": "supergroup",
    "title": "Herald Ops",
    "permissions": { "...": "..." }
  }
}
```

If you see `{"ok":false,"error_code":400,"description":"Bad Request: chat not found"}`, either the chat id is wrong-format (see §6 for the per-type shape) or the bot has not been added to the chat (see §2.3).

§4.3 Verify the pherald binary is reachable

```
go build -o /tmp/pherald ./pherald/cmd/pherald
/tmp/pherald version
```

Expected output:

```
pherald X.Y.Z (commit <sha>, built <date>)
```

If the binary builds but `version` panics, the workspace is broken (see `CLAUDE.md` § build + test). If the binary exits non-zero, file a bug — `version` is the simplest possible subcommand and must always work.

§4.4 Hermetic Send + persistence test

With `HERALD_TGRAM_BOT_TOKEN` + `HERALD_TGRAM_CHAT_ID` set, and a Postgres container running (see `docs/guides/OPERATOR_CREDENTIALS.md` § "Postgres" for the quickstart compose flow):

```
# Bring up Postgres if needed (one of these):
#   ./quickstart/up.sh                # full stack
#   bash containers/scripts/up.sh pg  # just Postgres

# Run the integration test — actually crosses the wire to
        Telegram:
go test -tags=integration -run TestSend_PersistsDeliveryEvidence
        \
        ./commons_messaging/channels/tgram/...
```

A PASS confirms: (a) the bot token is accepted by api.telegram.org; (b) the chat id resolves; (c) `sendMessage` returns a chat-side `message_id`; (d) the `outbound_delivery_evidence` row is persisted in Postgres with the correct evidence ceiling (`DeliveryRouted`); (e) the LIVE Bot API end-to-end works.

§4.5 Inbound smoke (`pherald listen`)

The end-to-end smoke: run `pherald listen` in the foreground, type a message in the chat, observe Claude Code dispatch + reply.

```
# Optional: capture the dispatch journal for §107.x evidence
mkdir -p docs/qa/HRD-100-smoke-$(date -u +%Y%m%dT%H%M%SZ)/

/tmp/pherald listen \
    --qa-out-dir docs/qa/HRD-100-smoke-$(date -u +
        %Y%m%dT%H%M%SZ) \
    2>&1 | tee /tmp/pherald-listen.log
```

Boot-time lines to look for in `/tmp/pherald-listen.log`:

```
pherald listen: starting Telegram getUpdates long-poll loop
pherald listen: bot.Me.Username=herald_ops_bot (self-filter active)
```

If `bot.Me.Username=` is empty, Subscribe will refuse to boot (see §8). Send a test message in the Telegram chat — within ~25s you should see a `<<<HERALD-DISPATCH-v1>>>` envelope being rendered to Claude Code in the log, followed by the bot's reply landing in your Telegram chat.

Tear down with `Ctrl-C` (graceful shutdown). Confirm the `--qa-out-dir` directory contains a `journal.jsonl` with one line per dispatch.

---

§5. Attachment size + type limits

The Bot API enforces strict hard caps that Herald inherits. Knowing them in advance avoids the silently-trimmed-or-rejected surprise at first prod use.

§5.1 Hard caps (uploads via the bot)

Type	Bot API endpoint	Hard limit	Notes
Text	`sendMessage`	**4096 chars**	UTF-16 code unit count, not byte count. MarkdownV2 escape characters count toward the limit. Herald does NOT auto-split — over-limit messages return `400 Bad Request: message is too long`.
Photo	`sendPhoto`	**10 MB** (free tier)	Photo uploads compress server-side; effective ceiling is ~10MB regardless of user account tier when the bot uploads.
Document	`sendDocument`	**50 MB**	The hardest cap. Holds even for premium users when uploaded BY THE BOT through the public Bot API. The workaround is the local Bot API server (`telegram-bot-api` self-hosted) — see §5.4.
Voice	`sendVoice`	**50 MB**	Same as document; voice messages are OGG/Opus encoded.
Video	`sendVideo`	**50 MB**	Same as document.
Audio	`sendAudio`	**50 MB**	Same as document.

§5.2 User-side download caps (inbound attachments)

When a subscriber sends a file to the bot, Herald reads it via `getFile` + the file-download endpoint. The bot can DOWNLOAD files up to **20 MB** through the standard Bot API. Larger files (which premium users can SEND in Telegram) trigger `Bad Request: file is too big` on `getFile`. This is a Bot-API-side limit; the workaround is again the local Bot API server.

§5.3 The content-addressed inbox

Every successfully-downloaded inbound attachment lands at:

```
${HERALD_INBOX_DIR:-$HOME/.herald/inbox}/tgram/<sha256-hex>.<ext>
```

The sha256 IS the filename stem; the extension is derived from the MIME via `channels.MimeToExt` (intentionally narrow — `.jpg`, `.png`, `.pdf`, `.ogg`, `.mp3`, `.mp4`, `.bin` fallback for unknown MIMEs). Idempotence: a duplicate download of identical bytes is a no-op (the existing file IS the proof of presence), so the long-poll never burns Bot API quota re-downloading the same file.

The streaming write helper is `channels.WriteContentAddressed("tgram", mime, body)` in `commons_messaging/channels/inbox.go`. It uses `io.MultiWriter(tempFile, sha256Hasher)` so the body flows to disk while the sha256 is computed inline; the final `os.Rename` is atomic, so partial writes are never visible at the content-addressed path. The temp file naming is `dl-*.part`; the next download starts fresh from a missing-temp state. This invariant is pinned by `TestDownloadAttachmentContentAddressed` in `commons_messaging/channels/tgram/attachments_test.go`.

§5.4 Lifting the 50 MB ceiling — local Bot API server

The 50 MB document upload cap and 20 MB download cap are properties of `api.telegram.org` (the public Bot API server). Telegram open-sourced the Bot API server itself; running a local instance against your own MTProto credentials raises the caps:

- **Document upload**: up to 2000 MB.
- **Document download**: full file size (Telegram's storage layer has no Bot API-level cap when the API server is local).

The local Bot API server is currently OUT OF SCOPE for Herald (HRD-NNN reserved). When it lands, Herald will route Bot API calls through a configurable `HERALD_TGRAM_BOT_API_URL` env var (default `https://api.telegram.org`).

§5.5 Where these limits surface in HRDs

HRD	Symptom
HRD-011	Outbound substrate — first to encounter the 50 MB cap during attachment fan-out tests.
HRD-100	Inbound runtime — first to encounter the 20 MB download cap on real subscriber-sent files.
HRD-134	Tgram-completeness — the structured rationale for NOT auto-splitting 4096+ char messages (the user must explicitly opt in to a splitter; silent splits are a §107 surprise).

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§6. Chat types — private vs group vs supergroup vs channel

Telegram has four chat types Herald cares about. Each has a different `chat_id` shape and a different permission model.

§6.1 Chat-type taxonomy

Type	`chat_id` shape	Example	Notes
`private`	positive integer, equal to the user's user_id	`123456789`	1:1 DM between the bot and exactly one Telegram user. Privacy mode is irrelevant — DMs are always fully visible.
`group`	negative integer (no specific prefix)	`-123456789`	"Basic group" — legacy format, max 200 members, no admin features. Privacy mode applies.
`supergroup`	negative integer starting with `-100`	`-1001234567890`	Modern group format, up to 200k members, full admin features. Privacy mode applies. Groups auto-convert to supergroups when admins are added.
`channel`	negative integer starting with `-100`	`-1009876543210`	One-way broadcast. The bot must be added as admin with `Post Messages` permission. `getUpdates` delivers `channel_post` updates (not `message`) — Herald's current inbound handler set

Type	`chat_id` shape	Example	Notes
			covers private/group/supergroup; channel inbound is reserved for future work.

§6.2 Where `chat_id` flows through Herald

1. **At boot**: `HERALD_TGRAM_CHAT_ID` is the default outbound destination — `Send(ctx, msg)` posts there unless `msg.Recipient.ChannelUserID` overrides it.
2. **Per inbound event**: `Subscribe`'s handler reads `update.Message.Chat.ID` and stamps it into `InboundEvent.Recipient.ChannelUserID` (decimal string form). The `channelRouter` in `pherald/internal/inbound/dispatcher.go` keys reply destination by this stamped value — replies go back to the chat the inbound came from.
3. **Per outbound reply**: `SendReplyGeneric(ctx, recipient, body, replyToID, attachments)` parses `recipient.ChannelUserID` back to `int64` for the Bot API call.

§6.3 Permission requirements

Chat type	Bot must be	To do
private	the other end of the DM	read + send messages (always, no extra permission)
group	member	send messages (always); read all messages only when privacy mode is disabled
supergroup	member	send messages (always); read all messages only when privacy mode is disabled OR the bot is an admin with `can_read_all_group_messages`
channel	admin with `Post Messages`	send messages (read-only inbound is not implemented for channels in Herald's current path)

If the bot is kicked from a chat or its admin scope is revoked, the next Bot API call to that chat returns `403 Forbidden: bot was kicked from …` — Herald surfaces this as a `subscriber-lost` signal (see §9.5).

§6.4 The `-100` prefix is real

A common operator confusion: when reading the chat id from `getUpdates`, the JSON shows `"id": -1001234567890` — the `-100` IS part of the canonical id, NOT a formatting artifact. Strip nothing. Set `HERALD_TGRAM_CHAT_ID='-1001234567890'` verbatim. Stripping the `-100` prefix gives you `1234567890` which Telegram resolves to a private chat with the corresponding user (if any), NOT your supergroup.

§6.5 Mapping to the spec

Spec V3 §43 catalogues the per-chat-type outbound routing matrix. Spec V3 §11.1 documents the Telegram-specific `Capabilities` (text + markdown + HTML + attachments + threads + interactive URL = true). Spec V3 §32.2 covers the inbound long-poll contract that drives the Wave 6 runtime.

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§7. Reply threading + `reply_to_message_id`

Telegram supports two distinct threading mechanisms; Herald uses one of them, intentionally.

§7.1 `reply_to_message_id` — quoted reply (used)

The classic Telegram reply: tap a message in the client, choose "Reply", type the reply. Telegram renders the reply with a small quote-block of the original above. The Bot API field is `reply_to_message_id` — pass the original message's integer id, and `sendMessage` produces the same quoted-reply visual.

Herald's `tgram.SendReply` (in `send.go`) uses this field:

```
opts := &telebot.SendOptions{
    ParseMode: parseMode,
    ReplyTo:   &telebot.Message{ID: replyToID},  // →
        reply_to_message_id in the wire form
}
sent, err := a.bot.Send(chat, text, opts)
```

The wire-byte assertion (`commons_messaging/channels/tgram/send_reply_test.go`) pins the exact form-value name (`reply_to_message_id`) so a refactor that dropped the field would surface immediately. Wave 6 mutation gate (M3 in the paired §1.1 suite) drops the assignment to prove the detector catches it.

§7.2 `message_thread_id` — forum topics (supported, V1 stub)

Supergroups can be enabled as "forums" — each top-level message starts a new topic that has its own thread id. The Bot API field `message_thread_id` routes a new message into a specific topic. Herald supports this via `OutboundMessage.Thread.ThreadID` (decimal string), threaded through to `telebot.SendOptions.ThreadID`:

```
if msg.Thread != nil && msg.Thread.ThreadID != "" {
    if tid, terr := strconv.Atoi(msg.Thread.ThreadID); terr ==
        nil {
        opts.ThreadID = tid
    }
}
```

The V1 implementation does not auto-discover topic ids — the caller must provide one. Auto-discovery (reading the `message_thread_id` from inbound updates and threading it back into the matching `SendReply`) is reserved for a future HRD.

§7.3 Attachment fan-out (the M3 rationale)

Telegram supports multi-attachment "albums" via `sendMediaGroup` — up to 10 photos/videos in one wire call. Herald deliberately does NOT use albums. The Wave 6.5 mutation gate (M3) pins the rationale:

- **Reply threading**: an album appears in the chat as ONE bundle with one quoted-reply pointer. If you reply to it, the reply quotes the bundle, not any individual attachment. Herald's inbound dispatcher loses the per-attachment correlation under albums.

- **Mixed-MIME bundles**: `sendMediaGroup` requires every member of the bundle to be the same media type (all photos, all videos, all audio, OR all documents — never mixed). Herald's outbound messages can carry mixed-MIME attachments (a doc + a photo + a voice memo), so album-based fan-out would force MIME-bucketing AND multiple album calls — defeating the latency advantage.
- **Error granularity**: an album call that fails fails for ALL its attachments. The serialised-per-attachment path Herald uses gives per-attachment success/failure.

So `SendReplyGeneric` posts the text reply first (carrying `reply_to_message_id`), then sends each attachment as a separate `sendDocument` / `sendPhoto` / `sendVoice` reply at the same thread depth — each carrying its own `reply_to_message_id` pointing at the same original message. The visual in the Telegram client is a series of quoted replies forming a tidy "reply chain" under the original.

§7.4 The native vs generic SendReply

Two methods exist for historical reasons:

Method	Signature	When to use
`Adapter.SendReply`	`(ctx context.Context, chatID int64, body string, replyToID int, attachments []commons.Attachment) (int, error)`	Direct Telegram callers (typed ids).
`Adapter.SendReplyGeneric`	`(ctx context.Context, recipient commons.Recipient, body, replyToID string, attachments []commons.Attachment) (string, error)`	Channel-agnostic callers (`channels.Channel` interface).

The native method predates the Wave 7 channel interface; `SendReplyGeneric` is a thin adapter that parses the decimal string ids into typed forms and calls the native method. See `tgram.go:227..243`.

---

§8. Self-filter — preventing echo loops

Telegram bots are bidirectional in the same channel: they BOTH send and receive in the same chat. Without active filtering, every reply Herald posts would arrive back through `getUpdates` as a fresh subscriber message, be dispatched to Claude Code, generate another reply, be dispatched again — an exponential loop that saturates the 25-second long-poll within the first round-trip.

§8.1 The defence

`pherald listen` (specifically `tgram.Adapter.Subscribe` in `subscribe.go`) caches the bot's own `@username` ONCE at Subscribe boot:

```
bot, err := telebot.NewBot(telebot.Settings{
    Token:  a.botToken,
    Poller: &telebot.LongPoller{Timeout: 25 * time.Second},
})
// telebot.NewBot dispatches getMe synchronously, populating
       bot.Me.
selfUsername := ""
if bot.Me != nil {
    selfUsername = bot.Me.Username
}
if selfUsername == "" {
    return fmt.Errorf("tgram.Subscribe: bot.Me.Username unset
       after NewBot — getMe likely failed; refusing to boot
       without self-filter (echo-loop hazard)")
}
self := channels.SelfIdentity{Kind: channels.IdentityUsername,
       Value: selfUsername}
```

Then EVERY inbound message is checked against this cached identity via the channel-agnostic filter `channels.IsSelfEcho` (Wave 7 T4 — see `MESSENGER_CHANNELS.md` §6 for the cross-channel contract):

```
ev := buildInboundEvent(msg)
if stampAndIsSelfEcho(ev, msg, self) {
    return nil  // drop — self-echo
}
return h.Handle(ev)
```

The mutation gate (Wave 6 T12 M1) pins this with a paired §1.1 test: removing the self-filter check causes the gate to mutate → assert-FAIL → restore. A live operator can never accidentally ship a self-filter-less binary.

§8.2 Fail-loud on empty identity

If `bot.Me.Username` is empty after `NewBot` (because `getMe` returned a degenerate user record, or because `telebot.Offline` was somehow enabled), `Subscribe` REFUSES to boot. There is NO `--allow-empty-identity` knob, NO `--skip-self-filter` knob, NO graceful-degradation path. A self-filter-less listener is a §107 PASS-bluff that LOOKS green but exponentially burns the operator's Claude quota on the first reply — the only safe behaviour is hard refusal.

§8.3 What if you legitimately need multi-bot conversation?

The filter is deliberately narrow: ONLY messages from THIS bot's own user record are dropped. A DIFFERENT bot in the same chat is kept and dispatched normally. So if you want bot-to-bot orchestration (a Slack bot pinging Herald's Telegram bot, etc.), this works out of the box — only the recursive self-loop is broken.

However, **bot-to-bot direct group messages are blocked at the Telegram API layer**, not the Herald layer. The memory file `telegram_bot_to_bot_wall` captures this: Telegram bots NEVER see another bot's group messages via `getUpdates`. A second-bot QA harness inside

Telegram is impossible at the Bot API tier — it would require an MTProto user-client (gotd/td-based). The QA workaround Herald uses is `qaherald`'s dedicated user-account-driven scenario harness (see `qaherald/cmd/qaherald/main.go`).

§8.4 Where to look in the code

- `commons_messaging/channels/tgram/subscribe.go:38..46` — the legacy `shouldDropBotSelf` helper (kept byte-for-byte because the Wave 6 M1 paired-mutation gate anchors its exact regex here).
- `commons_messaging/channels/tgram/subscribe.go:54..62` — `stampAndIsSelfEcho` — the live filter, used by every inbound handler.
- `commons_messaging/channels/selffilter.go` — the channel-agnostic comparator `IsSelfEcho`.
- `commons_messaging/channels/tgram/subscribe_test.go` — `TestSubscribeBotSelfFilter` pins the exact-text drop semantics.

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§9. Troubleshooting cookbook

§9.1 `401 Unauthorized`

Symptom: every Bot API call returns `{"ok":false,"error_code":401,"description":"Unauthorized"}`.

Root cause: token wrong, token revoked, or `HERALD_TGRAM_BOT_TOKEN` truncated by a quoting bug in `.env`.

Fix:

1. Confirm the env var is set in the running process — `cat /proc/$(pidof pherald)/environ | tr '\0' '\n' | grep HERALD_TGRAM` (Linux) or `ps eww -p $(pgrep pherald)` (macOS).
2. Re-fetch the token from BotFather: `/mybots → <bot> → API Token`.
3. If the token has been compromised, ROTATE via BotFather `/revoke`. There is no recovery — the old token is now invalid even if you reset the env var.
4. Restart `pherald listen`.

§9.2 `400 Bad Request: chat not found`

Symptom: `sendMessage` returns `{"ok":false,"error_code":400,"description":"Bad Request: chat not found"}`.

Root cause: chat id is wrong-format (e.g. `1234567890` for a supergroup that needs `-1001234567890`), or the bot has not been added to the target chat.

Fix:

1. Verify the chat id format matches §6.1.
2. Confirm the bot is a member of the chat: `curl -s "https://api.telegram.org/bot$TOKEN/getChat?chat_id=$ID" | jq` — if it returns `Bad Request: chat not found`, the bot is not in the chat (or `chat_id` is wrong).

3. For groups/supergroups/channels: add the bot via §2.3.

§9.3 `429 Too Many Requests`

Symptom: Bot API call returns `{"ok":false,"error_code":429,"description":"Too Many Requests: retry after N","parameters":{"retry_after":N}}`.

Root cause: Telegram rate-limiter triggered. Bot API has both global (`getUpdates`-side) and per-chat (`sendMessage`-side) rate caps.

Fix:

1. The `retry_after` field is the canonical signal — sleep for at least that many seconds before retrying.
2. Herald's adapter does NOT auto-backoff on 429 (yet — reserved HRD); a caller burst that hits 429 will surface the error. A future HRD will add exponential backoff with `retry_after` honour.
3. If you see persistent 429s, reduce send concurrency or add intentional spacing between sends to the same chat.

§9.4 `413 Request Entity Too Large` / `400 file is too big`

Symptom: `sendDocument` returns `413` (or `400 file is too big` for inbound `getFile`).

Root cause: exceeded the 50 MB upload cap (or 20 MB download cap) — see §5.

Fix:

1. Split the file (split + transmit as multiple messages).
2. Compress (zstd, xz, gzip) before upload.
3. For documents > 50 MB: run a local Bot API server (§5.4) — reserved HRD.

§9.5 `403 Forbidden: bot was kicked from the group chat`

Symptom: outbound to a previously-working chat returns `{"ok":false,"error_code":403,"description":"Forbidden: bot was kicked from the group chat"}`.

Root cause: a chat admin removed the bot, or the chat was deleted, or the user (in a DM) blocked the bot.

Fix:

1. This is the canonical "subscriber lost" signal — Herald should mark the subscriber inactive in the registry and stop attempting delivery.
2. To recover: re-add the bot via §2.3.
3. There is no automatic re-subscribe — by design, an explicit operator action is required to mark the subscriber active again.

§9.6 Echo loop — bot replying to itself

Symptom: the bot replies once, then a second, third, fourth reply arrives within seconds; the long-poll never quiesces.

Root cause: `bot.Me.Username` was empty at Subscribe boot OR the self-filter check was bypassed by a regression.

Fix:

1. Confirm Subscribe REFUSED to boot if `bot.Me.Username` was empty — if it booted with empty username, file a SECURITY bug (the §107 fail-loud invariant was broken).
2. Verify the live username: `curl -s "https://api.telegram.org/bot$TOKEN/getMe" | jq '.result.username'` — must be non-empty.
3. Grep your `pherald` binary's logs for the `bot.Me.Username=` boot line — confirm it carries your bot's canonical username (without `@`).
4. If you can reproduce in dev, run with race detector + log every dispatch — the loop will be visible within a few seconds.

§9.7 Long-poll stuck — no updates for >25s

Symptom: `pherald listen` is running but no inbound messages flow even when you send them in the chat.

Root causes (in descending order of likelihood):

1. **Privacy mode is ON in a group** (§2.2). Disable via BotFather `/setprivacy`.
2. **The bot is not in the chat you think it is** (§9.2 — silent variant; `getUpdates` returns empty rather than erroring).
3. **Network firewall to api.telegram.org:443**. Test: `curl -v https://api.telegram.org/bot$TOKEN/getMe`. If it hangs, your network blocks Telegram (corporate proxies are the usual suspect).
4. **A webhook is configured** — `getUpdates` returns updates only if NO webhook is set. Delete via `curl -X POST https://api.telegram.org/bot$TOKEN/deleteWebhook`. See §9.10 below.

§9.8 `409 Conflict: terminated by other getUpdates request`

Symptom: `pherald listen` boots, then aborts with `Conflict: terminated by other getUpdates request; make sure that only one bot instance is running`.

Root cause: Telegram allows ONLY ONE `getUpdates` consumer per token at a time. A second `pherald listen` instance, a forgotten dev shell, a CI job running the integration test, or a third-party debug client polling with your token will trigger this.

Fix:

1. Find the competing consumer: `pgrep -fa 'pherald listen'`.
2. Kill it: `pkill -f 'pherald listen'`.
3. Wait a few seconds (Telegram closes the conflicting session) and restart your `pherald listen`.
4. If you also have a webhook configured for the same token, `getUpdates` will conflict with it — `deleteWebhook` first (§9.10).

§9.9 Privacy mode — bot only sees commands in groups

Symptom: DMs work fine; group messages never arrive at the handler EXCEPT commands like `/foo` or messages that mention `@your_bot`.

Root cause: privacy mode is ON (the default for new bots).

Fix: DM `@BotFather`, `/setprivacy`, select your bot, `Disable`. Restart `pherald listen`. The bot will now see every message in groups it is a member of.

§9.10 Webhook conflicts with long-poll

Symptom: `getUpdates` returns the same updates repeatedly OR returns empty + `409 Conflict` arrives without a competing process.

Root cause: a webhook was previously configured on this token. Bot API does NOT permit `getUpdates` while a webhook is active.

Fix:

```
curl -s "https://api.telegram.org/bot${HERALD_TGRAM_BOT_TOKEN}/
        getWebhookInfo" | jq
```

If the `url` field is non-empty, delete the webhook:

```
curl -X POST "https://api.telegram.org/bot$
        {HERALD_TGRAM_BOT_TOKEN}/deleteWebhook"
```

Confirm with another `getWebhookInfo` — `url` must be empty. Then restart `pherald listen`.

§9.11 Zero-byte / `.part` residue in the inbox

Symptom: `~/.herald/inbox/tgram/dl-*.part` files remain after a download, or content-addressed final files are zero-byte.

Root cause: an inbound attachment stream was interrupted before `os.Rename` completed (network drop, process kill, disk full).

Fix:

1. Delete the `.part` files — they are deliberate temp residue; the next download starts fresh.
2. Zero-byte FINAL files would indicate a regression in `DownloadAttachment` — file a bug (the `attachments_test.go` invariant should have caught it pre-merge).

§9.12 Attachment ends up in `slack/` subdir (wrong inbox)

Symptom: a Telegram-delivered photo lands at `~/.herald/inbox/slack/<sha>.jpg` instead of `~/.herald/inbox/tgram/<sha>.jpg`.

Root cause: a refactor caused `Adapter.Name()` to drift from the registered channel name OR `HERALD_INBOX_DIR` is set to a path that overlaps another deployment.

Fix: see `MESSENGER_CHANNELS.md` §9.5.

§10. Pre-deploy operator audit

Run through this checklist before every fresh deploy or after rotating credentials.

§10.1 Environment

- [ ] `HERALD_TGRAM_BOT_TOKEN` is exported in the running pherald's environment. Format matches `<numeric-id>:<alphanumeric-hmac>`.
- [ ] `HERALD_TGRAM_CHAT_ID` is exported. Format matches §6.1 for the chat type (positive for DM, negative for group, `-100…` for supergroup/channel).
- [ ] `HERALD_INBOX_DIR` is either unset (default `~/.herald/inbox`) or points at a writable directory with at least worst-case-burst disk headroom (50 MB × expected concurrent uploads).
- [ ] No tokens appear in `git ls-files | xargs grep -l '<your-token-prefix>'` (manual spot-check).

§10.2 Bot configuration

- [ ] `/setprivacy` is `Disable` IF the bot is in groups and is meant to see all messages.
- [ ] Bot is a member of every target chat (`curl -s "https://api.telegram.org/bot$TOKEN/getChat?chat_id=$ID" | jq` returns `ok: true`).
- [ ] No competing `getUpdates` consumer is running (`pgrep -fa 'pherald listen'` returns at most one match).
- [ ] No webhook is configured (`curl -s "$/getWebhookInfo" | jq '.result.url'` returns `""`).

§10.3 Registry resolution

- [ ] `HERALD_CHANNELS` either is unset (defaults to `tgram`) or contains `tgram` as one of the comma-separated values.
- [ ] `grep 'channels.Register' commons_messaging/channels/*/*.go` lists `tgram` (sanity-check the blank-import is intact).

§10.4 Filesystem

- [ ] `~/.herald/inbox/` exists with mode `0700`.
- [ ]

- [ ] `~/.herald/inbox/tgram/` either exists with `0700` or will be auto-created on first attachment.
- [ ] Disk has worst-case-burst headroom (50 MB × concurrent uploads × 2 for safety).

§10.5 §107.x evidence

- [ ] Plan the run-id directory: `docs/qa/HRD-<NNN>-<TS>Z/` (timestamp UTC).
- [ ] For `pherald listen` smoke runs, pass `--qa-out-dir docs/qa/HRD-NNN-<TS>Z/` so the journal lands in the artefact directory.
- [ ] For manual operator-driven runs, screenshot both directions of the chat (subscriber message + bot reply) and commit the screenshots in the run-id directory.
- [ ] For `qaherald`-driven QA runs (Wave 5+), the transcript + attachments tree auto-lands under `docs/qa/qaherald-<TS>/` — confirm presence before tagging.

§10.6 First-message smoke

- [ ] Build the binary: `go build -o /tmp/pherald ./pherald/cmd/pherald`.
- [ ] Boot: `/tmp/pherald listen --qa-out-dir docs/qa/HRD-NNN-<TS>Z/ 2>&1 | tee /tmp/pherald-listen.log`.
- [ ] Boot line `pherald listen: bot.Me.Username=…` shows non-empty username (self-filter active).
- [ ] Type a one-word test message in the target chat (e.g. `hello`). Within ~25s, a reply arrives in the chat.
- [ ] The reply quotes the original (`reply_to_message_id` set).
- [ ] `docs/herald/diary/main.md` contains the new conversation entries (in + out).
- [ ] `docs/qa/HRD-NNN-<TS>Z/journal.jsonl` contains one line per dispatch.
- [ ] Ctrl-C gracefully shuts down (no zombie process).

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§11. References

§11.1 Spec V3

- **§11.1** — Telegram channel `Capabilities` declaration (text + markdown + HTML + attachments + threads + interactive URL = true; `AttachmentMaxMiB = 50`; `DeliveryCeiling = DeliveryRouted`).

- **§32.2** — inbound long-poll contract; `getUpdates` timeout = 25s; observational 30s safety-net timer.
- **§32.9** — anti-echo-loop self-filter mandate (the §107 binding for `bot.Me.Username` capture + `IsSelfEcho` fan-out).
- **§43** — channel + flavor catalogue; per-chat-type routing matrix.
- **§107.x** — `docs/qa/<run-id>/` evidence mandate (Helix §11.4.83 cascade).

§11.2 HRDs

HRD	Wave	Status	Description
HRD-011	1	LIVE	Telegram outbound substrate — `Send`, MarkdownV2, attachments, persistence.
HRD-100	6	in_progress (awaiting T10b live evidence)	`pherald listen` inbound runtime — `getUpdates` long-poll, Claude Code dispatch, `<<<HERALD-REPLY>>>` action router.
HRD-101	6	LIVE	Wave 6 paired §1.1 mutation gate covering inbound runtime (M1=self-filter, M2=envelope, M3=attachment fan-out).
HRD-110	7	LIVE	Extract `channels.Channel` interface (richer inbound contract embedding §11.0).
HRD-111	7	LIVE	`init()`-based channel registry.
HRD-112	7	LIVE	Per-channel inbox subdirs (`~/.herald/inbox/tgram/`).
HRD-113	7	LIVE	Generalize bot self-filter via `BotSelfIdentity` + `IsSelfEcho`.
HRD-114	7	LIVE	Multi-channel `pherald listen` fan-in.
HRD-133	tgram-completeness	open	Telegram capabilities matrix completeness review.
HRD-134	tgram-completeness	open	Rationale for not auto-splitting 4096+ char messages.
HRD-135	tgram-completeness	open	Local Bot API server support (lifts 50 MB cap).
HRD-136	tgram-completeness	open	Forum-topic auto-threading (inbound `message_thread_id` → outbound `SendReply`).
HRD-137	tgram-completeness	open	Channel-post inbound handler (`update.channel_post`).
HRD-138	tgram-completeness	open	This guide.

§11.3 Source files

- `commons_messaging/channels/tgram/tgram.go` — `Adapter`, `New` / `NewWithCreds` / `NewWithStorage`, `init()` registry binding, `BotSelfIdentity` / `SendReplyGeneric` / `DownloadAttachment` (channel interface methods).
- `commons_messaging/channels/tgram/send.go` — `Send` (outbound), native `SendReply(chatID int64, replyToID int)`.

- `commons_messaging/channels/tgram/subscribe.go` — `Subscribe` long-poll loop, `stampAndIsSelfEcho` filter, OnText/OnPhoto/OnDocument/OnVoice handlers.
- `commons_messaging/channels/tgram/attachments.go` — `DownloadAttachment(ctx, bot, fileID, mime)`.
- `commons_messaging/channels/tgram/healthcheck.go` — `HealthCheck` (calls `getMe`).
- `commons_messaging/channels/tgram/persist.go` — `outbound_delivery_evidence` persistence.
- `commons_messaging/channels/channel.go` — the eight-method channel interface.
- `commons_messaging/channels/selffilter.go` — `StampSender`, `IsSelfEcho` (channel-agnostic).
- `commons_messaging/channels/inbox.go` — `WriteContentAddressed`, `InboxDir`, `MimeToExt`.
- `pherald/cmd/pherald/listen.go` — `loadEnabledChannels`, `perChannelConfig`, `runListen`, `channelRouter`.
- `pherald/internal/inbound/dispatcher.go` — Claude Code dispatch + reply-action router.

§11.4 Plan + design docs

- `docs/superpowers/plans/2026-05-21-wave6-cc-headless-bridge.md` — Wave 6 inbound runtime plan.
- `docs/superpowers/plans/2026-05-27-wave7-generic-messenger.md` — Wave 7 multi-channel framework.
- `docs/CONTINUATION.md` §3 — live-test handoff prompt (operator-supplied credentials).

§11.5 Sibling guides

- `MESSENGER_CHANNELS.md` — multi-channel framework, registry semantics, multi-channel `pherald listen` semantics, `HERALD_CHANNELS` env var.
- `OPERATOR_CREDENTIALS.md` — umbrella credentials guide; the `.env` resolution order; the audit checklist.
- `messengers/TELEGRAM.md` — the original HRD-011 setup walkthrough (Step 1..Step 7) — more verbose for first-time operators; this guide is the operator-deep-dive complement.
- `messengers/SLACK.md` — Slack-specific deep-dive (HRD-115).
- `HERALD_CONSTITUTION.md` — §107 end-user-usability covenant + §107.x evidence mandate.
- `dispatchers/CLAUDE_CODE.md` — Claude Code dispatcher (HRD-012) — the other half of the inbound round-trip.

§11.6 Constitutional anchors

- Helix Universal Constitution §11.0 — outbound `Channel` contract.
- Helix Universal Constitution §11.4 / §11.4.4 / §11.4.5 / §11.4.6 — anti-bluff captured-evidence mandate.
- Helix Universal Constitution §11.4.83 — `docs/qa/<run-id>/` evidence cascade (Herald §107.x).
- Helix Universal Constitution §11.4.85 — stress + chaos test mandate.
- Herald §107 — end-user-usability covenant.

- Herald §107.x — docs/qa evidence mandate.
- Herald §107.y — working-tree quiescence rule.

§11.7 External

- Telegram Bot API documentation — canonical reference for every endpoint Herald calls (`getMe`, `getUpdates`, `getChat`, `getFile`, `sendMessage`, `sendPhoto`, `sendDocument`, `sendVoice`, `deleteWebhook`, `getWebhookInfo`).
- BotFather — bot creation + management.
- @RawDataBot — third-party chat-id helper bot (use with caution; vet before sharing sensitive content).
- Telegram Bot API server (open source) — the local server that lifts the 50 MB cap (HRD-135 reserved).

Sources verified

Per HelixConstitution §11.4.99 + Herald §108.n (Latest-Source Documentation Cross-Reference Mandate). Every operator-facing instruction in this document was cross-referenced against the LATEST official online documentation of the relevant service before publication.

Last verified: 2026-05-28

Source	URL / path	Authored / verified
Telegram official Bot API documentation	https://core.telegram.org/bots/api	§2 (BotFather walkthrough — `/newbot`, `/setprivacy`, `/setname`, `/setdescription`, `/setuserpic`, `/setcommands`, `/deletebot`, `/revoke`); §3 (token format `<bot-id>:<api-token>`); §4 (`getMe`, `getChat`, `getUpdates`, `getWebhookInfo`, `deleteWebhook`); §5 (sendMessage 4096-char limit, sendPhoto/Document/Voice/Video/Audio 50 MB upload cap, getFile 20 MB download cap); §6 (chat-type taxonomy: private positive id / group negative / supergroup `-100…` / channel `-100…`); §7 (reply_to_message_id quoted-reply, message_thread_id forum-topic, sendMediaGroup album restrictions); §8 (`getMe.result.username` for self-filter identity); §9 (401/400/403/409/413/429 error code semantics, `Bad Request: chat not found`, `terminated by other getUpdates request`); §9.10 (webhook ⊕ long-poll mutual exclusion).

Source	URL / path	Authored / verified
Telegram Bot Features — Privacy Mode	https://core.telegram.org/bots/features#privacy-mode	§2.2 (privacy-mode-ON default); §6.3 (`can_read_all_group_messages` flag semantics); §9.7 + §9.9 (privacy-mode-as-#1-stuck-cause); §10.2 (operator audit checklist entry).
Telegram Bot API local server (open source)	https://github.com/tdlib/telegram-bot-api	§5.4 (lifting the 50 MB cap via self-hosted Bot API server — 2000 MB document upload, full-size download); §5.5 (HRD-135 reserved).
telebot.v3 vendored library	`submodules/telebot/` (v3.3.8, pinned per §11.4.74)	§7 (`telebot.SendOptions.ReplyTo` → `reply_to_message_id`, `telebot.SendOptions.ThreadID` → `message_thread_id`); §8 (`telebot.NewBot` synchronous `getMe` populating `bot.Me`); §11.3 source-file mapping.
Empirical Herald operator testing 2026-05-28	`docs/qa/HRD-LIVE-20260528T082128Z/`	§1.2 status (Wave 6 + Wave 7 LIVE assertions); §4.4 hermetic Send-evidence test path; §4.5 inbound smoke envelope shape.
HelixConstitution §11.4.99 (this document's authority)	`<parent>/constitution/Constitution.md` §11.4.99 (HelixConstitution commit `c640947`)	This footer (pattern + cadence requirement).

Re-verification cadence (per §11.4.99 (C)): Telegram is a risk-classified service (per §11.4.99 (D) and Herald §108.n (D) — bot APIs face automated-bot-detection + anti-abuse-system bans). Telegram-specific sections (every section §2..§9 of this document) → **90-day max staleness**, next re-verification due **2026-08-26**. Re-verify earlier on: Bot API breaking-change announcement (the `bots/api` changelog at https://core.telegram.org/bots/api#recent-changes), operator-error reports (a captured `description` field that does not match the cookbook), Herald vN.0.0 release boundary.

End of guide.